SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	**Total Marks:**	100
CO2 Statement:	*Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)*		
Student Name:	ATHIN SIVA.S	**Reg. No.:**	192524237
Section / Batch:	2025	**Date:**	27/7/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q61

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I _ATHIN SIVA.S__________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _________________________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department

Name:	Name:	Name:
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Q61. Closest Pair – Warehouse Location Analysis

Aim

To identify the two nearest warehouses using the Brute Force Closest Pair Algorithm, compute the distance between every pair of warehouses, count the total pairwise comparisons, and analyze the algorithm's efficiency.

Problem Interpretation

A logistics company has warehouses located at the following coordinates:

(2,1), (5,4), (6,5), (10,9), (3,2), (8,7)

The objective is to determine the two warehouses that are closest to each other. Finding the nearest warehouses helps in reducing transportation costs, improving inventory sharing, and enabling faster emergency deliveries.

Why Brute Force?

- It checks every possible pair of warehouses.
- It guarantees the correct minimum distance.
- It is easy to implement.
- It is suitable for a small number of warehouse locations.

Formula Used

The Euclidean distance between two points is

$$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

Step 1: Label the Warehouses

Warehouse	Coordinates
W1	(2,1)
W2	(5,4)
W3	(6,5)
W4	(10,9)
W5	(3,2)
W6	(8,7)

Step 2: Compute Distances Between Every Pair

$W_1 - W_2 = \sqrt{(5-2)^2+(4-1)^2} = \sqrt{9+9} = \sqrt{18} = 4.24$

$W_1 - W_3 = \sqrt{(6-2)^2+(5-1)^2} = \sqrt{16+16} = \sqrt{32} = 5.66$

$W_1 - W_4 = \sqrt{(10-2)^2+(9-1)^2} = \sqrt{64+64} = \sqrt{128} = 11.31$

$W_1 - W_5 = \sqrt{(3-2)^2+(2-1)^2} = \sqrt{1+1} = \sqrt{2} = 1.41$

$W_1 - W_6 = \sqrt{(8-2)^2+(7-1)^2} = \sqrt{36+36} = \sqrt{72} = 8.49$

$W_2 - W_3 = \sqrt{(6-5)^2+(5-4)^2} = \sqrt{1+1} = \sqrt{2} = 1.41$

$W_2 - W_4 = \sqrt{(10-5)^2+(9-4)^2} = \sqrt{25+25} = \sqrt{50} = 7.07$

$W_2 - W_5 = \sqrt{(3-5)^2+(2-4)^2} = \sqrt{4+4} = \sqrt{8} = 2.83$

$W_2 - W_6 = \sqrt{(8-5)^2+(7-4)^2} = \sqrt{9+9} = \sqrt{18} = 4.24$

$W_3 - W_4 = \sqrt{(10-6)^2+(9-5)^2} = \sqrt{16+16} = \sqrt{32} = 5.66$

$W_3 - W_5 = \sqrt{(3-6)^2+(2-5)^2} = \sqrt{9+9} = \sqrt{18} = 4.24$

$W_3 - W_6 = \sqrt{(8-6)^2+(7-5)^2} = \sqrt{4+4} = \sqrt{8} = 2.83$

$W_4 - W_5 = \sqrt{(3-10)^2+(2-9)^2} = \sqrt{49+49} = \sqrt{98} = 9.90$

$W_4 - W_6 = \sqrt{(8-10)^2+(7-9)^2} = \sqrt{4+4} = \sqrt{8} = 2.83$

$W_5 - W_6 = \sqrt{(8-3)^2+(7-2)^2} = \sqrt{25+25} = \sqrt{50} = 7.07$

Step 3: Distance Summary Table

Pair	Distance
W1 - W2	4.24
W1 - W3	5.66
W1 - W4	11.31
W1 - W5	1.41
W1 - W6	8.49
W2 - W3	1.41
W2 - W4	7.07
W2 - W5	2.83
W2 - W6	4.24
W3 - W4	5.66
W3 - W5	4.24
W3 - W6	2.83
W4 - W5	9.90
W4 - W6	2.83
W5 - W6	7.07

Step 4: Closest Pair

Minimum distance: $\sqrt{2} \approx 1.41$

There are two closest pairs:

- W_1 (2,1) and W_5 (3,2)
- W_2 (5,4) and W_3 (6,5)

Both have the same minimum distance of 1.41 units.

Step 5: Number of Pairwise Comparisons

For $n = 6$ warehouses,

$$\frac{n(n-1)}{2} = \frac{6\times5}{2} = 15$$

Total comparisons = 15

Time Complexity Analysis

Case	Time Complexity
Best Case	$O(n^2)$
Average Case	$O(n^2)$
Worst Case	$O(n^2)$

Since every pair must be examined, the running time remains $O(n^2)$ in all cases.

Correctness Verification

✓ All possible warehouse pairs were examined.

✓ Euclidean distance was calculated for each pair.

✓ The smallest distance (1.41) was identified correctly.

✓ Therefore, the algorithm correctly finds the nearest warehouses.